

A Deep Learning-Based Thermal Prediction Approach for Energy Management in Cloud Data Centers

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Abstract—Escalating host temperatures in data centers can create hot spots, significantly boosting cooling costs and impacting reliability. Accurately predicting host temperatures is of the utmost importance in managing resources effectively. Existing temperature estimation solutions are inefficient due to a lack of accurate prediction. To this end, we proposed a deep learning-based thermal prediction approach called *DLTPA*, which aims to minimize the temperature and power consumption of virtual machines (VMs) in data centers. Specifically, we built a Long Short-Term Memory (LSTM) network model to accurately predict host temperature and power consumption, demonstrating superior performance over traditional algorithms. Our LSTM model exhibits exceptional accuracy in predicting thermal and energy dynamics, achieving an R^2 value of 0.98 in power consumption prediction, indicating exact forecasts. Furthermore, we design an efficient VM placement strategy to achieve host peak temperature reduction by rationally arranging VM tasks. The experimental results demonstrate that the *DLTPA* significantly improves over other leading-edge algorithms. It reduces peak power consumption by 3.64% to 9.39%, lowers average temperature by 3.21% to 7.96%, achieves a 0% SLA violation rate, and maintains a high level of load balancing at 19.8%.

Index Terms—Thermal Prediction, Power Consumption, VM Placement, Deep Learning, Thermal Management.

I. INTRODUCTION

Data centers are known for their high energy demands, and inefficient resource allocation can lead to unnecessary energy consumption. This raises operational costs and has a detrimental environmental impact, as data centers are significant contributors to global energy consumption and carbon emissions. In light of these challenges, there is a growing need for more intelligent and efficient VM placement policies that take both temperature and power consumption into account. Predictive models, such as Long Short-Term Memory (LSTM) networks, provide promising solutions by forecasting host CPU temperature and package power consumption based on historical data. By integrating well-trained predictive

models into optimization algorithms, data centers can make informed decisions that balance performance, cost, and sustainability.

Effective prediction of the host temperature of data centers is critical for maintaining hardware reliability and efficiency [1]. Overheating can cause hardware failures and shorten the lifespan of components, potentially leading to more severe system malfunctions. Traditional cooling solutions, while effective, tend to be energy-intensive and costly. By accurately predicting host CPU temperatures using advanced LSTM network models [2]–[4], thermal loads can be proactively managed across the data center. The trained predictive models are employed after stochastic VM placement to ensure adequate thermal control [5].

Package power consumption refers to the energy consumption of server racks or host enclosures. Power prediction models allow data centers to anticipate the energy needs of different workloads, enabling optimized resource allocation [6]. By integrating power consumption predictions into VM placement policies, data centers can effectively reduce wasted computing energy, lower cooling equipment costs, and ensure that host power consumption remains within reasonable limits [7].

This paper introduced a deep learning-based thermal prediction approach called *DLTPA* for energy management in cloud data centers. It is divided into two phases: the host temperature prediction phase and the VM placement phase. In the host temperature prediction phase, by employing a predictive model based on the LSTM algorithm, our approach predicts hosts' heat and power requirements, leading to more informed and efficient VM placement decisions. Through this work, we aim to contribute to the development of more intelligent and efficient resource management solutions in the field of cloud computing. The main contributions of this paper are enumerated as follows:

- 1) We introduced a thermal prediction approach

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called DLTPA for energy management in cloud data centers. This approach comprises two phases: host temperature prediction and VM task repositioning.

- 2) Our approach utilizes LSTM models to accurately predict host temperature and power consumption. These models leverage historical data to forecast future heat and energy demands, facilitating more informed and efficient VM placement decisions.
- 3) A hybrid algorithm based on Genetic Algorithm and Simulated Annealing for optimal VM allocation to reduce energy consumption and CPU temperature in data centers.
- 4) We have validated the predictive performance of the proposed DLTPA through extensive simulations and comparisons with other experimental approaches. The results demonstrate significant improvements in energy efficiency and heat management, highlighting the practical applicability of our approach in real-world cloud data centers.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III describes our research problem's formulation and the prediction model's development. In Section IV, we discuss the application of the prediction module to VM scheduling. The experimental results are presented in Section V. Finally, Section VI concludes the paper and outlines future research directions.

II. RELATED WORK

Predicting CPU temperature in data centers remains a challenging task, with current techniques often limited by accuracy or computational complexity. Effective thermal management not only reduces operational costs but also enhances system stability.

A. Host Temperature and Energy Consumption Prediction Techniques

Machine learning models have significantly advanced temperature and energy prediction, surpassing traditional statistical methods. Ilager *et al.* [8] proposed an XGBoost-based temperature prediction model using private cloud data. Nisce *et al.* [7] applied regression models and XGBoost to predict CPU temperature from CPU utilization and energy data. Wang *et al.* [9] introduced a thermal model using improved linear regression and multi-channel historical data. Zhang *et al.* [10] developed a neural network model to address performance degradation under uneven thermal conditions. Lu *et al.* [11] introduced ThermTransformer, a real-time spatial thermal mapping method. Chen *et al.* [12] developed L-PAW, a deep learning-based cloud workload prediction algorithm.

Most models focus on either temperature or energy consumption prediction. This paper addresses that gap by introducing an LSTM-based framework that integrates

both, supporting a hybrid GA-SA optimization algorithm for improved temperature management and energy efficiency in VM placement.

B. VM Placement Strategy

VM placement is a critical research area in cloud computing, focusing on optimizing VM distribution across physical hosts to enhance energy efficiency, thermal management, and load balancing.

Several VM placement algorithms have been proposed to achieve these goals. Mustafa *et al.* [7] introduced the Best-Fit Decreasing (BFD) algorithm to minimize resource wastage and energy consumption. Peake *et al.* [13] proposed a parallel ant colony optimization (PACO) method to reduce operational costs by leveraging modern processor technology. Ibrahim *et al.* [14] developed a power-aware Particle Swarm Optimization (PAPSO) technique to balance power consumption and SLA compliance. Karmakar *et al.* [15] applied ant colony optimization to reduce deployment and communication costs. Feng *et al.* [16] proposed a Simulated Annealing and Greedy (SAG) algorithm to reduce energy consumption while meeting SLAs.

While these algorithms have improved energy efficiency and communication latency, they often overlook host temperature, which is critical for system stability and hardware longevity. Addressing this gap is vital for modern data centers.

III. SYSTEM OVERVIEW

In this section, we design the system model and temperature and energy model followed by the problem statement targeted by our paper. Fig. 1 illustrates the overall architecture of our methodology. Thereinto, Y denotes the actual value, and $\hat{Y}$ is the predicted value. The model predicts temperature $\hat{T}$ and power consumption $\hat{P}$ based on the input features, optimized to minimize MSE and MAE while maximizing R^2 . The DLTPA algorithm processes the VM-list, incorporating $\hat{T}$ and $\hat{P}$, to produce an output sequence that reduces both temperature and power consumption. This framework provides an efficient solution for optimizing VM placement, lowering power consumption, preventing overheating, maintaining load balance, and ensuring SLA compliance.

A. System Model

As shown in Fig. 2, our system model integrates the placement and prediction modules to operate seamlessly. The core of this integration lies in the precise prediction algorithms that forecast hosts' power consumption and CPU temperature, laying the groundwork for optimal host placement decisions. The interaction between these modules helps minimize operational costs.

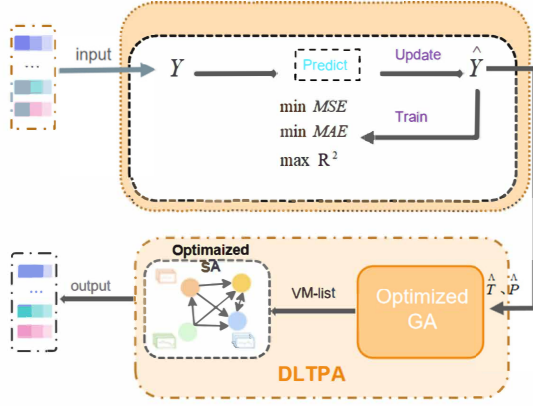


Fig. 1: DLTPA strategic architecture diagram.

The prediction module's data aggregator collects host data from the cloud data center, which is used to pre-train the prediction models. During pre-training, models with the highest accuracy are selected based on training metrics. These models are then deployed in the placement module, which retrieves relevant host characteristics. Initially, VMs are placed randomly, but subsequent placements are based on predicted power consumption, temperature, and other resource constraints.

VM placement decisions are driven by updates from the predictive model and changes in data center workloads. The prediction module continuously collects real-time data (e.g., CPU utilization, memory, network I/O) to update the model. Based on these updates, the system adjusts VM placement to align with current conditions, optimizing for reduced CPU temperatures and power consumption.

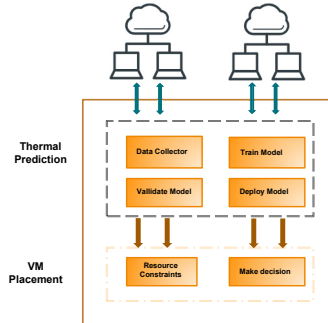


Fig. 2: A layout of the system

B. Temperature and Energy Model

To address the prediction task, we evaluated various machine learning algorithms [8], including LSTM, MLP, DT, SVM, Bagging, RNN, and a hybrid LSTM-CNN model, to identify the most accurate method for predicting host CPU temperature and power consumption. Since each host has dual-core CPUs, we averaged the

two temperature values to simplify the data, obtaining a single T_{CPU} value. The dataset was organized by host ID, with each host's dataset comprising variable-length tuples D_{in} , detailed in Table I, to optimize host-level predictions.

TABLE I: Attributes in D_{in} for Host-level Metrics

Symbol	Description
CPU	CPU utilization (%)
R	RAM usage (MB)
R_x	Available RAM (MB)
N_{CPU}	Number of CPU cores
N_{CPUx}	Used CPU cores
N_{Rx}	Inbound network traffic (Kbps)
N_{Tx}	Outbound network traffic (Kbps)
N_{vm}	Number of VMs
P_c	Power consumption (watts)
T_{CPU}	CPU temperature ($^{\circ}\text{C}$)

Based on this analysis, we developed temperature and power consumption prediction models. The following optimization strategy balances energy efficiency and thermal management to optimize VM placement in cloud data centers:

$$\begin{aligned}
 & \min \left(\max_{i=1}^N T_i, \sum_{i=1}^N P_i \right) \\
 & \text{subject to } T_i \leq T_{\text{threshold}}, \quad \forall i = 1, 2, \dots, N \\
 & P_i \leq P_{\text{max}}, \quad \forall i = 1, 2, \dots, N \\
 & \sum_{j=1}^{V_i} \text{CPU}_{ij} \leq \text{CPU}_i, \quad \forall i = 1, 2, \dots, N \\
 & \sum_{j=1}^{V_i} \text{Mem}_{ij} \leq \text{Mem}_i, \quad \forall i = 1, 2, \dots, N
 \end{aligned} \tag{1}$$

where T_i represents the predicted peak temperature of host i , while P_i denotes its predicted power consumption. N is the total number of hosts, CPU_{ij} is the CPU usage of VM j on host i , and Mem_{ij} is the memory usage of VM j on host i , ensuring that total resource usage does not exceed the available CPU cores and memory on each host.

IV. ALGORITHM DESIGN OF DLTPA

A. Learning with LSTM

LSTM, a deep learning algorithm introduced by Schmidhuber *et al.* [17], processes and predicts sequential data. By incorporating gating mechanisms, LSTM effectively addresses the vanishing and exploding gradient problems commonly faced by traditional Recurrent Neural Networks (RNNs) when handling long sequences [18]. The LSTM model utilizes three gates: the forget gate, input gate, and output gate, allowing it to capture essential features across various time scales, thereby achieving accurate time-series predictions [19], [20].

In the field of deep learning technology, LSTM is a key tool for analyzing sequential data and is widely applied in time-series forecasting tasks. A typical LSTM model is defined as:

$$\begin{aligned}
f_t &= \sigma(W_{fh}h_{t-1} + W_{fx}x_t + b_f) \\
i_t &= \sigma(W_{ih}h_{t-1} + W_{ix}x_t + b_i) \\
\tilde{c}_t &= \tanh(W_{ch}h_{t-1} + W_{cx}x_t + b_c) \\
c_t &= f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \\
o_t &= \sigma(W_{oh}h_{t-1} + W_{ox}x_t + b_o) \\
h_t &= o_t \cdot \tanh(c_t)
\end{aligned} \tag{2}$$

where f_t , i_t , and o_t represent the outputs of the forget, input, and output gates at time step t ; c_t is the cell state at time step t ; x_t is the input vector; $W_{fh}, W_{fx}, W_{ih}, W_{ix}, W_{ch}, W_{cx}, W_{oh}, W_{ox}$ are the weight matrices associated with hidden states and inputs for each gate and the cell state; b_f, b_i, b_c, b_o are the biases; and h_t represents the hidden state. The functions $\tanh()$ and $\sigma()$ are the hyperbolic tangent and sigmoid activation functions, respectively.

Temperature prediction has vital applications across various domains, enabling efficient cooling system operation [21], detecting thermal anomalies [22], [23], and optimizing resource management [24], [25]. We developed a prediction model based on historical host data, integrated into our placement algorithm. Using sensor-collected data, the model forecasts host CPU temperature and power consumption in real time during the placement process, leveraging host characteristics for robust data support.

B. Heuristic Algorithms Tackle VM Placement

Empirical data reveals an inverse relationship between package power consumption and CPU temperature. Due to the dynamic nature of data center operations, these two objectives often conflict. Reducing power does not necessarily lower peak temperature, as different components (e.g., CPU, memory, I/O) affect temperature differently. Similarly, lowering CPU temperature through techniques like DVFS or cooling mechanisms may not reduce overall power consumption. Thus, balancing power consumption and peak temperature is critical. We model the VM placement problem as a multi-objective optimization task, aiming to minimize both package power consumption and peak temperature while keeping them within safe thresholds. Our approach provides theoretical and practical frameworks to enhance data center stability and efficiency.

We propose a heuristic optimization approach combining Genetic Algorithm (GA) with Simulated Annealing (SA). GA identifies a near-optimal solution, which SA fine-tunes by exploring the solution space. Starting from GA's best solution, SA adjusts placements, accepting better solutions and occasionally worse ones based on

a decreasing temperature parameter, allowing the algorithm to escape local optima. This process continues until convergence, improving energy management and temperature control.

Details of the heuristic algorithm are shown in **Algorithm 1**. A population is initialized by randomly placing VMs, and fitness is evaluated based on predicted temperature and energy consumption:

$$\begin{aligned}
\text{fitness} &= u_1 \times \text{totalTemp} + u_2 \times \text{totalEnergy} \\
u_1 + u_2 &= 1
\end{aligned} \tag{3}$$

Experimental results show the best performance when $u_1 = 0.6$ and $u_2 = 0.4$, where totalTemp and totalEnergy represent the predicted temperature and energy of the data center for a given configuration. GA selects the top 50% of individuals from the population and applies crossover and mutation to generate a new population iteratively for a fixed number of generations. Subsequently, Simulated Annealing (SA) is applied for further optimization. SA explores neighboring placements by perturbing the current solution and calculating the change in energy (ΔE). A new solution is accepted if $\Delta E < 0$, or with a probability of $\exp(-\Delta E/T)$ if $\Delta E \geq 0$, where T is the temperature parameter. The temperature T decreases iteratively until convergence, leading to an optimized VM placement and updated host configurations.

The heuristic optimization approach proposed combines a Genetic Algorithm (GA) and Simulated Annealing (SA). It has a time complexity of $O(G * P * (H + M)) + O(I * H)$, where G represents the number of generations, P is the population size, H is the number of hosts, M is the number of VMs, and I is the number of iterations for SA. The GA dominates the overall complexity due to its population-based approach, while the SA provides further refinement with less computational cost. This balanced approach ensures efficient VM placement and optimizes temperature and energy management in cloud data centers.

V. EXPERIMENT RESULTS

A. Experimental Setup

To train the model for predicting host temperature and power consumption, we utilized a carefully curated dataset¹ from a private data center. The dataset includes features such as CPU utilization, memory usage, network I/O, and host temperature. Temperature and power consumption were predicted based on correlations with key variables, including CPU and memory usage, network traffic, and the number of virtual machines. HWINFO² was deployed across all hosts to collect data, stored in

¹https://github.com/Lynsana/Train_data

²<https://www.hwinfo.com>

Algorithm 1 Optimize VM Placement using GA and SA

```

1: Input: VMs, Hosts, Mean/Std Dev for Temperature
   and Energy, Prediction Models (nettemp, netenergy)
2: Output: Updated Hosts, Total Temperature, Total
   Energy, Predicted Temperature, Predicted Energy
3: Initialize: Population size = 50, generations = 100,
   mutation rate = 0.1,  $u_1 = 0.6$ ,  $u_2 = 0.4$ 
4: for each generation do
5:   for each individual in population do
6:     Evaluate fitness based on predicted tempera-
       ture and energy:
7:      $fitness = u_1 \cdot totalTemp + u_2 \cdot totalEnergy$ 
8:   end for
9:   Select top 50% individuals and apply crossover
     and mutation to generate a new population
10: end for
11: Simulated Annealing: Set  $T = 100$ ,  $T_{min} = 0.8$ ,
     $\alpha = 0.9$ 
12: while  $T > T_{min}$  do
13:   Perturb solution to explore new placements
14:   Calculate  $\Delta E$  between new and current place-
     ments
15:   if  $\Delta E < 0$  or  $e^{-\Delta E/T} > rand()$  then
16:     Accept new solution
17:   end if
18:   Update temperature:  $T = \alpha \cdot T$ 
19: end while
20: Return: Optimized VM placement and updated
    hosts

```

CSV format. For preprocessing, we used the pandas³ library in Python to clean the dataset and remove invalid measurements, ensuring data quality.

The experiments were conducted in a cloud simulation environment developed on the MATLAB platform, with all algorithms implemented using MATLAB's machine learning library. Default parameters were used for each algorithm. For MLP, we constructed a 3-layer network using MATLAB's Neural Network Toolbox, with five neurons in the hidden layer, one neuron in the output layer, and the ReLU activation function.

During the placement algorithm phase, we leveraged the real-world Bitbrain dataset [26], capturing resource consumption metrics from critical business workloads. These metrics aligned with the features in our pre-training data, enabling accurate prediction within the placement algorithm.

We compared our approach against three well-known algorithms: TAS [8], MoVPAAC [27], and Round Robin (RR). TAS focuses on reducing peak host temperature, MoVPAAC addresses multi-objective VM placement with an emphasis on power reduction, and RR distributes VMs evenly across hosts. These algorithms provided a

³<https://github.com/pandas-dev/pandas.git>

solid foundation for benchmarking the performance of our proposed method.

B. Performance Evaluation

In our experiments, conducted within a cloud data center environment, we considered the following metrics that influence host CPU temperature and power consumption:

To evaluate the performance of the algorithms, we used three metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2). The Mean Absolute Error (MAE) is a metric used to evaluate the performance of prediction models, particularly in regression analysis. It is defined as follows:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

where y_i is the i -th actual value, $\hat{y}_i$ is the i -th predicted value, and n is the total number of data points. A smaller MAE indicates that the model's predictions are closer to the actual values.

The Mean Squared Error (MSE) is another metric used to assess the performance of predictive models, which evaluates the accuracy of predictions by calculating the average squared difference between the predicted and actual values. A lower MSE indicates higher prediction accuracy. It is defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (5)$$

The R-squared (R^2) is a statistical metric that measures the goodness of fit for a regression model. It indicates the proportion of the variance in the dependent variable that the model explains. The formula for R^2 is:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (6)$$

where SS_{res} (Residual Sum of Squares) represents the sum of squared differences between the predicted and actual values, and SS_{tot} (Total Sum of Squares) represents the total variation in the dependent variable. A higher R^2 value suggests a better fit of the model.

As shown in Fig. 3, the evaluation metrics for each algorithm are illustrated. From Fig. 3a and Fig. 3b, it is clear that the LSTM algorithm consistently achieves lower MAE and MSE values, indicating superior accuracy in predicting both host temperatures and power consumption. Additionally, Fig. 3c demonstrates that LSTM exhibits a significantly higher R^2 value, further confirming its exceptional predictive performance. Given the strong performance of the LSTM algorithm across these metrics, with high accuracy and low error rates, it forms a solid foundation for subsequent scheduling strategy development.

We employed predictive models to calculate the temperature and power usage for the TAS, MOVPAAC, and

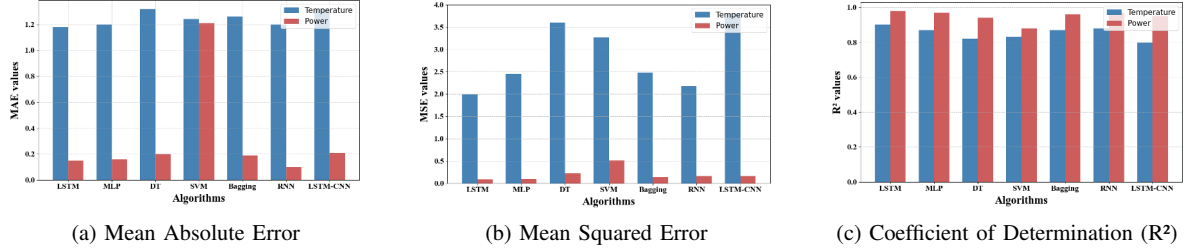


Fig. 3: Performance evaluation metrics of each predictive algorithm

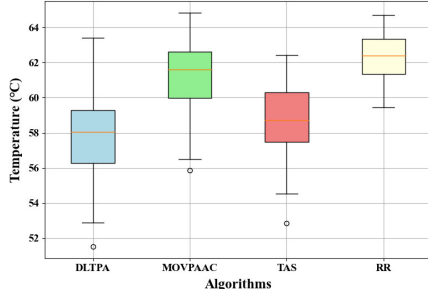


Fig. 4: Temperature distribution analysis

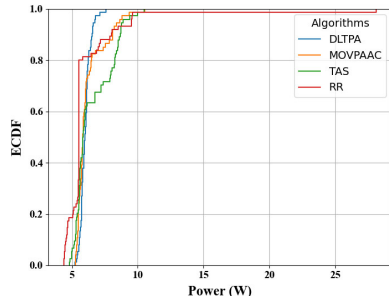


Fig. 5: Power distribution density

RR algorithms. Fig. 4 shows the average temperature of all hosts across the four algorithms during each scheduling interval. The box plot highlights that hosts running the DLTPA algorithm achieve the lowest median temperature, while the RR algorithm exhibits the highest median, with a more balanced temperature range. This analysis provides valuable insights for optimizing the performance of temperature and power consumption algorithms.

Fig. 5 depicts the power consumption density plot for all hosts. It shows that DLTPA's power consumption distribution is more concentrated, with lower dispersion, indicating that power usage is within a narrower range. In contrast, TAS shows a broader power consumption distribution, with the highest values, suggesting more high-power data points and a more scattered distribution.

Fig. 6 presents the power consumption data across algorithms. MoVPAAC achieves the lowest power con-

sumption through Ant Colony Optimization, while our algorithm, though not as efficient in reducing power, demonstrates stable performance. TAS, focusing solely on temperature reduction, leads to the highest power consumption.

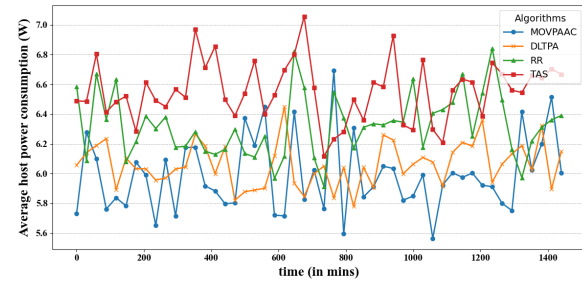


Fig. 6: Average Power in each scheduling interval

Fig. 7 presents the host temperatures under each algorithm. The results demonstrate that DLTPA is the most effective at reducing host temperatures, as it prioritizes temperature control over power consumption, achieving superior cooling performance. In contrast, the round-robin (RR) algorithm yields the highest temperatures due to evenly distributed VMs, causing resource contention on certain hosts with higher resource demands. MoVPAAC, while focused on power reduction, inadvertently raises host CPU temperatures by reducing the cooling system workload.

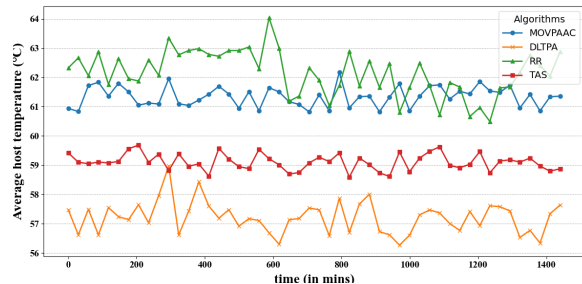


Fig. 7: Average Temperature in each scheduling interval

Load Balance (LB): Optimal load balancing aims to distribute CPU and memory resources across hosts

evenly. A higher standard deviation indicates greater disparity in resource usage, while a lower value reflects a more balanced system. The goal is to achieve uniform resource utilization across hosts, ensuring efficient operation and smooth system performance.

$$LB = \sqrt{(\sigma_{\text{cpuUtilization}})^2 + (\sigma_{\text{memoryUtilization}})^2} \quad (7)$$

where $\sigma_{\text{cpuUtilization}}$ is the standard deviation of CPU utilization, $\sigma_{\text{memoryUtilization}}$ is the standard deviation of memory utilization.

SLA Violation Rate: The SLA violation rate indicates if the hosts are meeting the Service Level Agreement (SLA) for temperature and energy consumption. An SLA violation occurs when the predicted temperature or power consumption exceeds the maximum values specified in the agreement, namely T_{max} and P_{max} . A high SLA violation rate suggests instability and a failure to meet expected performance levels, while a lower violation rate indicates better adherence to the SLA.

In our experiments, based on the collected observations from the test set, we set T_{max} to 75°C and P_{max} to 30W.

$$\text{SLA Violation Rate} = \frac{\text{slaViolations}}{\text{totalSLA}} \quad (8)$$

where `slaViolations` represents the number of hosts that exceeded the maximum temperature or power consumption, and `totalSLA` represents the total number of hosts involved in the SLA calculation.

As shown in Fig. 8, RR achieves the lowest load balance index by distributing tasks evenly across hosts without accounting for the current load, while our algorithm, considering resource constraints, achieves comparable performance. TAS, focused solely on temperature reduction, leads to imbalances by overloading certain hosts. In Fig. 9, DLTPA records a 0% SLA violation rate due to its balanced optimization of temperature and power consumption. In contrast, algorithms like MoVPAAC, though effective in reducing power, fail to meet SLA requirements, resulting in violations.

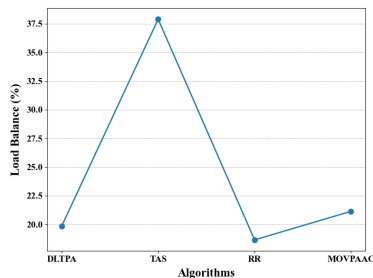


Fig. 8: Average Power in each scheduling interval

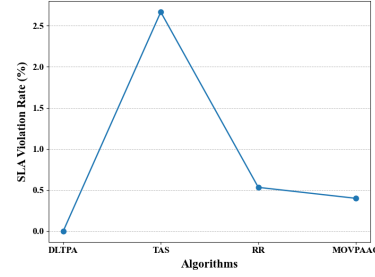


Fig. 9: Average Power in each scheduling interval

VI. CONCLUSION

In our research, we introduce DLTPA, a deep learning-based thermal prediction approach for VM placement in cloud data centers. We conducted an evaluation of seven machine learning algorithms: LSTM, MLP, DT, SVM, Bagging, RNN, and LSTM-CNN, to make thermal-aware predictions. Our findings show that LSTM achieved the best prediction accuracy, with close to 98% accuracy for power prediction and 90% accuracy for temperature prediction. Additionally, we propose an enhanced heuristic algorithm, GA-SA, which combines the genetic algorithm and simulated annealing algorithm. The main objective of this algorithm is to reduce power consumption and temperature in cloud data centers. GA generates an optimal solution, which is further explored by SA to adjust VM placements within the solution space.

To validate the performance of DLTPA, we compared our DLTPA with TAS, MOVPAAC, and RR. The experimental results indicate that our DLTPA obviously decreases the temperature and power consumption of a host's CPU, under the premise of keeping up with the SLA requirement with effective load balancing. Specifically, DLTPA may reduce peak power consumption by 3.64% to 9.39% and average temperature by 3.21% to 7.96%, making the SLA violation rate equal to 0% and keeping the load balance over 19.8% in high-level.

In the future, we will extend our approach in two directions. First, we will explore multi-objective optimization further to incorporate more goals toward overall energy efficiency improvement. Secondly, we plan to extend the predictions to GPU or PUE metrics by exploring more comprehensive and layered prediction strategies.

ACKNOWLEDGMENT

This work is supported by the National Natural Science Foundation of China under Grant No. 62072214, the specific research fund of The Innovation Platform for Academicians of Hainan Province under Grant No.YSPTZX202410, the South China Normal University teacher research and cultivation fund under Grant No.KJF120240001, the South China Normal University School of Artificial Intelligence Development Fund under Grant No.210123..

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